ADC + DAC: Md. Jahin Alam Jishnu Mahmud

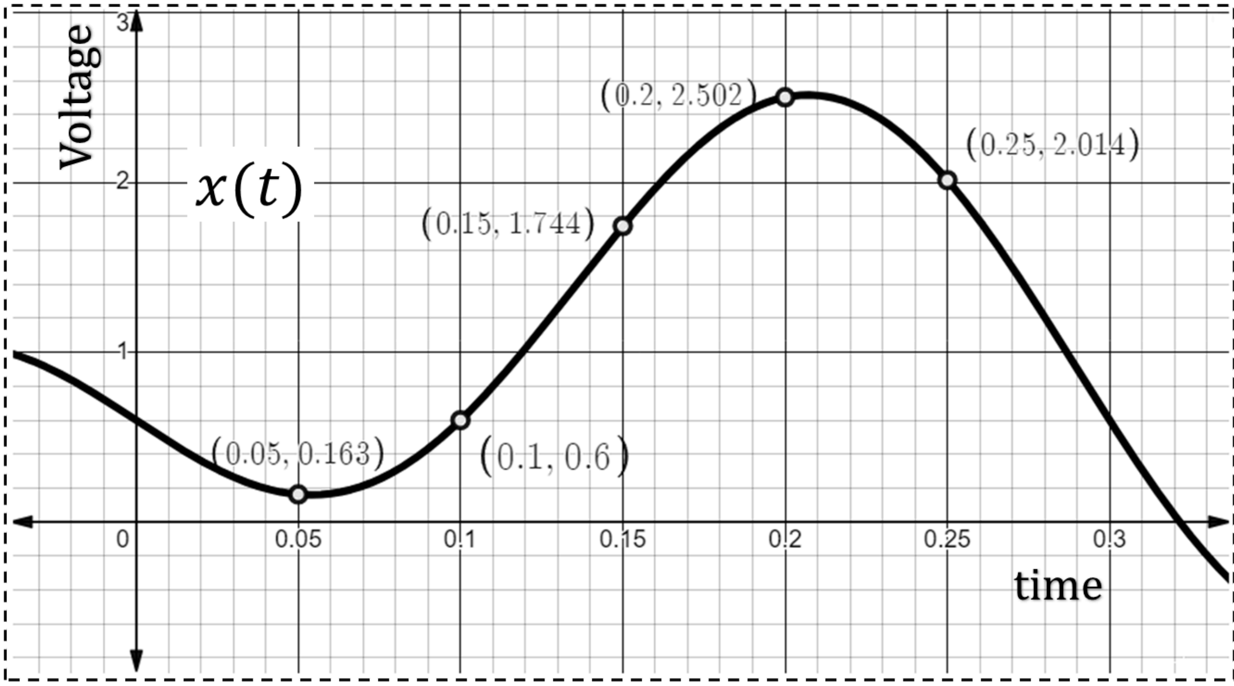
Signal Generator + Schmitt trigger: Md Obaidur Rahman, Himaddri Roy

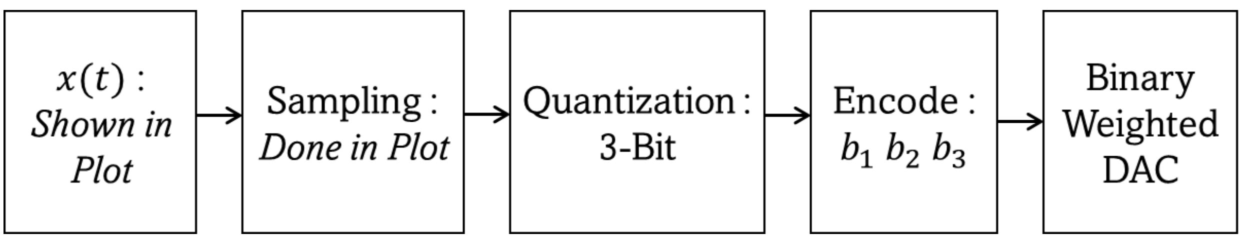
MOS+CMOS: Tasnim Ferdous apu, Quazi Shafayat Hossain

- While posting your question here, please follow this format:

(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

ADC+DAC[MJA]





A signal x(t) is provided. A process of 'AC-DC' and 'DC-AC' conversion is done to the signal. The plot shows x(t) and 5 sampled (not quantized) values. The entire process is shown in the flow chart above. The conversion is done using the following known quantities:

$$V_{min} = 0\,V,\quad V_{max} = 2.6\,V,\quad V_{REF}\ (for\ DAC) = 2.6\,V$$ [Set-A]

$$V_{min} = 0\,V,\quad V_{max} = 3.2\,V,\quad V_{REF}\ (for\ DAC) = 3.2\,V$$ [Set-B]

(a)	CO2	Determine the Resolution, Δ. Then, find all the quantization ranges for the 'AC-DC' conversion above.	[2+8]
(b)	CO2	Find the quantized ranges that will be assigned to the 5 sampled values. Also, find the encoded version for each of them [$b_1\ b_2\ b_3$].	[2.5+2.5]
(c)	CO2	Use the encoded values of the 5 points and pass it through the DAC shown above. What are the 5 outputs from it?	[5]

Set A :-

(a) $\Delta = \frac{2.6}{2^3} = 0.325V$

Quantization Range

7	2.275 – 2.6	111
6	1.95 – 2.275	110
5	1.625 – 1.95	101
4	1.3 – 1.625	100
3	0.975 – 1.3	011
2	0.65 – 0.975	010
1	0.325 – 0.65	001
0	0 – 0.325	000

(b) 5 sampled values

0.163	0.6	1.744	2.502	2.014
↓	↓	↓	↓	↓
0	1	5	7	6
↓	↓	↓	↓	↓
000	001	101	111	110

$\boxed{b_1 \; b_2 \; b_3}$

Set-B :-

(a) $\Delta = \frac{3.2}{2^3} = 0.4$

Quantization Range

7	2.8 – 3.2	111
6	2.4 – 2.8	110
5	2.0 – 2.4	101
4	1.6 – 2.0	100
3	1.2 – 1.6	011
2	0.8 – 1.2	010
1	0.4 – 0.8	001
0	0 – 0.4	000

(b) 5 sampled values

0.163	0.6	1.744	2.502	2.014
↓	↓	↓	↓	↓
0	1	4	6	5
↓	↓	↓	↓	↓
000	001	100	110	101

$\boxed{b_1 \; b_2 \; b_3}$

(c) ADC V_o [$R_f = R$]

$$= V_{Ref}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4}\right)$$

The 5 outputs : $V_{Ref} = 2.6$

(i) 0V
(ii) 0.65V
(iii) 3.25V
(iv) 4.55V
(v) 3.9V

Does not Match

(c) ADC V_o [$R_f = R$]

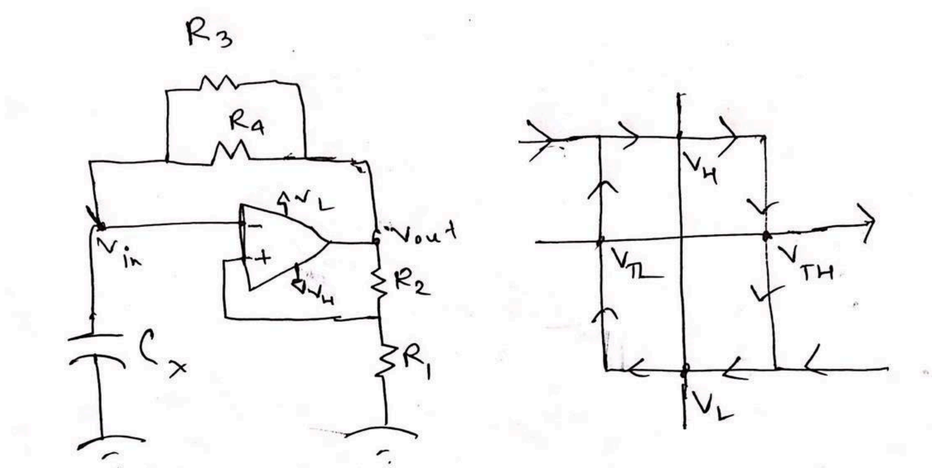
$$= V_{Ref}\left(b_1 + \frac{b_2}{2} + \frac{b_3}{4}\right)$$

The 5 outputs : $V_{Ref} = 3.2$

(i) 0V
(ii) 0.8V
(iii) 3.2V
(iv) 4.8V
(v) 4V

Does not Match

Schmitt Trigger[OBR]



For the following circuit, the value of Cx, R2, R3 and R4 are 0.05µF, 15kΩ , 60kΩ and 100 kΩ. In the figure, the values of V_H, V_L and V_{TH} are 8V, -10V and 5.5 V.

(a)	CO3	Find the hysteresis Width and value of R1.	[2.5+2.5]
(b)	CO3	Find the frequency and duty cycle of the signal generator.	[7]
(c)	CO3	Draw the graph of (Vin vs t) and (Vout vs t) with clearly showing the maximum and minimum value of each signal and time period.	[4]
(d)	CO3	If R3 is removed , what are the effects on duty cycle and time period? If V_{TH} is increased, what are the effects on duty cycle and time period?	[2+2]

Set-A

(a)

$$V_{TH} = \frac{R_1}{R_1+R_2} V_H$$

$$\Rightarrow \left(5.5 = \frac{R_1}{R_1+15} \times 8\right)$$

$$\Rightarrow \boxed{R_1 = 33k\Omega}$$

$$\frac{V_{TL}}{V_{TH}} = \frac{V_L}{V_H} = (-6.875V) \leftarrow V_{TL}$$

$$\therefore \text{width} = \boxed{12.375V}$$

(b) $R_x = (100 \| 60) = 37.5k\Omega$

$$T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$$

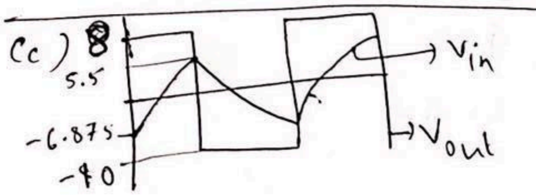
$$\boxed{T_1 = 3.34\,ms}$$

$$T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$$

$$\boxed{T_2 = 3\,ms}$$

$$f = \frac{1}{T_1+T_2} = \boxed{157.728 Hz}$$

(c)



(d) R_3 remove $\rightarrow R_x \uparrow \rightarrow T$ increase

duty cycle $\rightarrow$ fix

Set-B

(a)

$$V_{TL} = \frac{R_1}{R_1+R_2} V_L$$

$$\Rightarrow (-7.5) = \frac{R_1}{R_1+15} \times (-10)$$

$$\Rightarrow \boxed{R_1 = 45k\Omega}$$

$$\frac{V_{TH}}{V_{TL}} = \frac{V_H}{V_L} \Rightarrow \boxed{V_{TH} = 6V}$$

$$\therefore \text{width} = \boxed{13.5V}$$

(b) $R_x = (65 \| 90) = 37.74k\Omega$

$$T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$$

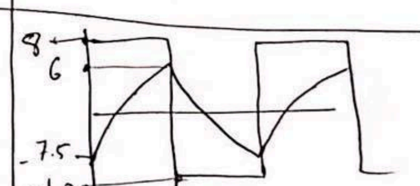
$$\boxed{T_1 = 3.47\,ms}$$

$$T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$$

$$\boxed{T_2 = 3.15\,ms}$$

$$f = \frac{1}{T_1+T_2} = \boxed{151.057 Hz}$$

(c)



(d) $\Rightarrow V_{TH} \uparrow \rightarrow T_1 \uparrow, T_2 \downarrow$

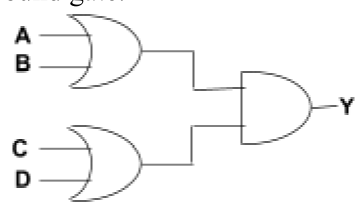
$T \rightarrow$ uncertain

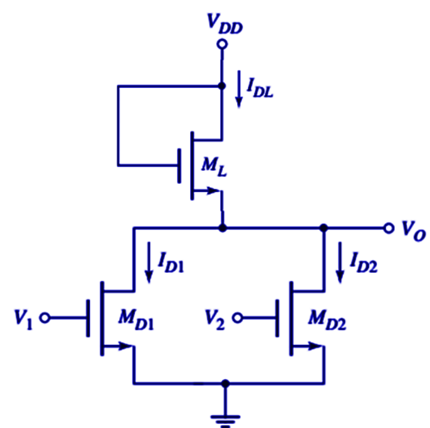
duty cycle $\rightarrow \uparrow$.

MOSFET+CMOS[TAF]

Ques1:

The enhancement-load NMOS NOR gate in the figure is biased at V_{DD} = 5 V. The transistor parameters are K_n = 0.3mA/V^2 , V_{TN1} = 0.5 V, V_{TN2} = 0.8 V, $V_{TN L}$ = 0.7 V.

CO1	(a)	Find the operating mode of the load NMOS transistor.	[2]
CO1	(b)	Find the value of v_o **in V and I_{DL}, I_{D1}, I_{D2} in *mA*** for v_1= **5V**, v_2= **5V**	[4+6]
CO1	(c)	Given is a static CMOS compound gate.  **i. Identify** the boolean function for output, Y. **ii. Design** the CMOS circuit for Y. Clearly show the PMOS and NMOS networks.	[2+6]



Solution :

setA: $K_n = 0.3 mA/V^2$, $V_{TN1} = 0.5$, $V_{TN2} = 0.8V$; $V_{TNL} = 0.7V$

setB: $K_n = 0.35 mA/V^2$, $V_{TN1} = 0.55$, $V_{TN2} = 0.8V$; $V_{TNL} = 0.7V$

a) $V_{DS} \geq V_{GS} - V_{TH} \rightarrow$ [saturation] $V_{GS_L} = V_{DSL}$ [operating mode → ML → $\boxed{\text{2 marks}}$

As, $V_{DS_L} > V_{DSL} - V_{TNL}$; M_L always in saturation.

b) $V_1 = V_2 = 5$ V   [V_0, I_{DL}, I_{D1}, I_{D2} → $\boxed{\text{2 marks each}}$]

$I_{DL} = I_{D1} + I_{D2}$

$$\Rightarrow K_n (V_{GS_L} - V_{TNL})^2 = K_n\left[2\cdot(V_{GS_{D1}} - V_{TND1}) V_{DS_{D1}} - V_{DS_{D1}}^2\right] + K_n\left[2\cdot(V_{GS_{D2}} - V_{TND2}) V_{DS_{D2}} - V_{DS_{D2}}^2\right]$$

$$\Rightarrow (V_{DD} - V_0 - V_{TNL})^2 = \left[2(V_1 - V_{TND1}) V_0 - V_0^2\right] + \left[2(V_2 - V_{TND2}) V_0 - V_0^2\right]$$

$\Rightarrow$

setA

$\Rightarrow (4.3 - V_0)^2 = 17.4 V_0 - 2V_0^2$

$\Rightarrow 18.49 - 8.6V_0 + V_0^2 = 17.4V_0 - 2V_0^2$

$\Rightarrow 3V_0^2 - 26V_0 + 18.49 = 0$

$\Rightarrow V_0 = 7.885$ × $> V_{DD}$

① $V_0 = \boxed{0.781\ V}$

② $I_{D1} = 3.71$ mA

③ $I_{DD1} = 1.925$ mA

④ $I_{DD2} = 1.785$ mA

setB

$\Rightarrow (4.3 - V_0)^2 = 17.3V_0 - 2V_0^2$

$\Rightarrow 18.49 - 8.6V_0 + V_0^2 = 17.4V_0 - 2V_0^2$

$\Rightarrow 3V_0^2 - 25.9V_0 + 18.49 = 0$

$\Rightarrow V_0 = 7.847$ × $> V_{DD}$

① $V_0 = \boxed{0.785\ V}$

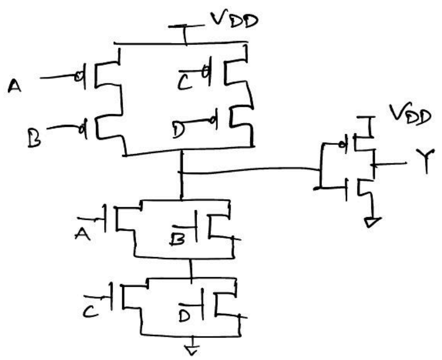
② $I_{D1} = 4.32$ mA

③ $I_{DD1} = 2.229$ mA

④ $I_{DD2} = 2.092$ mA

c) $Y = (A+B)\cdot(C+D)$ [2 marks → Boolean Expression]

[4+4 marks → PMOS + NMOS]



Dual Slope[JMD]

Solution

Dual slope analog-digital solution:

Set A #3

(a) $t_1 = 4 \times 10^{-3}$ analog value $= 2 + 2\sin(200\pi t_1) = 3.18$ V

$V_{ref} = 4$V

$\left\{ \frac{\text{Rad}}{3.18} \text{ ✓} \quad \frac{\text{Deg}}{2.09} \text{ allow} \right.$

$\frac{3.18}{4.00} = \frac{m}{2^N}$ $N = 4$ bit

$m = 12.72\text{V} \longrightarrow \boxed{1100}$

(b) $T_1 = T_p \times 2^N$

$T_1 = \frac{1}{2 \times 10^6} \times 2^4 = 8 \times 10^{-6}$ s

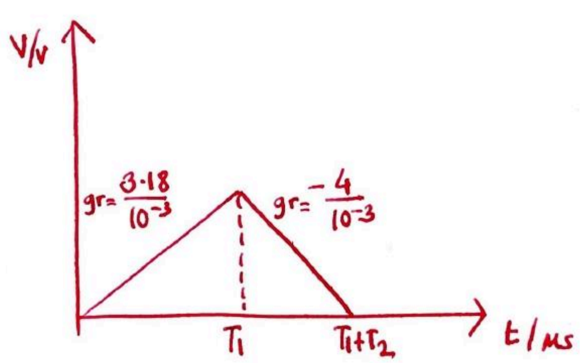
$T_2 = T_p \times m$

$T_2 = 12.72 \times \frac{1}{2 \times 10^6} = 6.36 \times 10^{-6}$ s

$\left[\text{allow } T_2 = 12 \times \frac{1}{2 \times 10^6} = 6 \times 10^{-6}\text{s} \right]$

total time $= T_1 + T_2 = (8 + 6.36) \times 10^{-6} \boxed{= 14.36 \times 10^{-6}\text{s}}$

$\left[\text{allow } T_1 + T_2 = (8+6) \times 10^{-6} = 14 \times 10^{-6}\text{s} \right]$



$RC = 1k\Omega \cdot 1\mu F$
$= 10^3 \cdot 10^{-6} = 10^{-3}$

(c) $T_{max} = 2T_1 = 2 \times 8 \times 10^{-6} = 16 \times 10^{-6}$s

$f_{max\ possible} = \frac{1}{16 \times 10^{-6}} = 62{,}500$ Hz

$F_s = 4000$ Hz

$f_{max\ possible} > f_{s\ sampled}$ } ADC can support this rate

CSE 350

Section : B

#2

(a) analog value $y_1 = 2 + 2\cos(300\pi t)$ $t_1 = 5\times10^{-3}s$

rad

analog value = 2V

Degree (allow)

= 3.993 V

$V_{ref} = 4V$

$$\frac{m}{2^4} = \frac{2}{4} \qquad m = 8 \longrightarrow \boxed{1000}$$

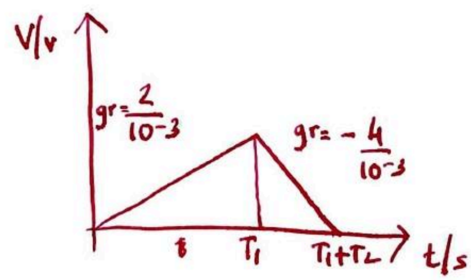
(b) $T_1 = T_p \times 2^N = \frac{1}{2\times10^6} \times 2^4 = 8\times10^{-6}s$ $T_2 = T_p \times m$

$T_2 = 8 \times \frac{1}{2\times10^6} = 4\times10^{-6}s$

total time $= T_1 + T_2 = (8\times10^{-6} + 4\times10^{-6}) \boxed{= 12\times10^{-6}s}$

$RC = 1k\Omega \cdot 1\mu F$

$= 10^3 \cdot 10^{-6} = 10^{-3}$



(c) $T_{max} = 2T_1 = 16\times10^{-6}s$

$f_{max\ possible} = \frac{1}{16\times10^{-6}} = 62{,}500Hz$

$F_s = 5000Hz$

$f_{max\ possible} > f_{s\ sampled}$

ADC can support this rate